

REASONING ARTIFACT

The Dimensional Invariant:

Sleep Duration, Sleep Topology, Attractor Basin Depth, and Long-Term Survival Across the Entire History of Life

The oldest surviving lineages sleep the least.

The minimum of the sleep spectrum is the maximum of the survival spectrum.

The topology of sleep payment is itself a geometric prediction of the landscape.

This is not a coincidence. This is the geometry.

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DERIVATION NOTICE

The hypothesis in this document was derived from first principles using the attractor geometry framework ($S + N + G \rightarrow R$) before literature consultation. The derivation emerged from the sleep-as-basin-navigation-cost principle and the sleep sonar instrument (companion documents, same repository, 2026-03-18), and was extended to a universal claim about deep-time survival before any palaeontological or comparative survival literature was consulted.

This version (v2.0) extends v1.0 by making explicit a geometric claim that was implicit in the original derivation but not fully stated: the **topology of the sleep payment process** — the architecture by which an organism discharges its basin-occupation cost — is itself a geometric prediction of the framework, not a measurement artefact or calibration problem. Different landscape positions do not only produce different sleep durations. They produce different sleep *topologies*. This is a stronger and more complete claim. The crocodilian data (Kelly & Lesku, 2015, 2019) makes the topological dimension empirically explicit and is recorded here as confirmation of a predicted geometric property, not as an anomaly requiring special treatment.

All claims were confirmed against literature after derivation. The derivation record is the priority-establishing document.

THE DIMENSIONAL INVARIANT

Across all life, at all scales, across all of deep time: the total cost of an organism's attractor basin position must be discharged through rest. The framework makes two inseparable predictions:

Prediction 1 — Quantity: The total cost discharged per cycle (sleep duration or rest-state equivalent) tracks the navigational demand of the organism's landscape position. Basin depth predicts survival horizon. The oldest surviving lineages carry the lowest navigational cost. They discharge the least.

Prediction 2 — Topology: The architecture by which cost is discharged — the sleep topology — is itself a geometric property of the landscape position, not an independent biological variable. Different landscape positions do not merely require different amounts of sleep. They require different *kinds* of sleep. The topology is predicted by the geometry.

The oldest surviving lineages sleep the least. How they sleep is the geometry of where they are.

1. The Framework

1.1. The Attractor Geometry Framework

$$S + N + G \rightarrow R$$

- S = structural substrate (genome and encoded build capacity)
- N = coherence gradient field (external or internal field specifying basin occupation)
- G = global attractor landscape (basin depths, barrier heights, accessible states, topology of transitions)
- R = resulting phenotype or state

The framework predicts that an organism's position in G determines three things simultaneously:

1. The total navigational cost of occupying that position — what must be discharged.
2. The survival trajectory — how resistant the basin is to perturbation.
3. The topology of the discharge process — what architecture of rest is geometrically required to pay the cost from that position.

These three are not independent. They are three projections of a single geometric object: the organism's position in the landscape G .

1.2. The Core Claim: Cost, Topology, and Survival as a Single Geometric Object

The metabolic cost of occupying a position in an attractor landscape is a function of the navigational demand imposed by that position. Navigational demand is determined by: (1) the number of accessible basins; (2) the height of barriers between basins; (3) the frequency of required transitions; (4) the depth of Basin Ω if one exists; and (5) the within-basin metabolic or computational intensity of the occupied basin.

Rest is the generic process by which this accumulated cost is discharged. **The architecture of that rest process — its topology — is not a separate biological variable. It is a further projection of the same landscape geometry.**

An organism in a single deep basin with zero navigational demand has no synaptic reconfiguration cost to discharge. Its rest approaches the minimum and its topology approaches the simplest possible: metabolic quiescence or biochemical circadian rhythm only.

An organism in a symmetric dual-basin landscape discharging high between-basin switching cost requires consolidated bilateral deep sleep to perform the full synaptic reconfiguration that context switching demands. The topology must be bilateral because the cost is bilateral — both hemispheres participated in the navigation and both must be reset.

An organism whose ecological landscape position is ecologically generalist but whose metabolic architecture is ectothermic faces a topological constraint: the metabolic cost of full bilateral wakefulness exceeds its energetic budget for most of the day. The geometric solution is a distributed intermediate rest state — bilateral in EEG signature (because the navigational cost is moderate and not requiring hemispheric asymmetry) but extended across time rather than concentrated into a single deep-sleep window. This is not an anomaly. It is the landscape geometry expressing itself through the only metabolically feasible topology available to an ectothermic Navigator.

The dolphin faces a different but structurally equivalent constraint: the cost of full bilateral sleep in an open ocean environment is mortality. The landscape geometry resolves this by dissolving the barrier between waking and sleep entirely — the two basins merge into a permanent intermediate state, Basin Ω of the waking/sleep landscape itself, maintained via unihemispheric EEG alternation.

These are not special cases requiring individual explanation. They are the framework's topological predictions working correctly.

Sleep topology is the geometry of where you are. Duration is how much the geometry costs. Survival is how deep the geometry goes.

2. The Sleep Cost Axes

The total cost discharged through sleep is decomposable into three axes. Each axis corresponds to a distinct geometric feature of the landscape position.

1. **Axis 1 — Between-basin switching cost.** The organism navigates between multiple accessible basins under its own motivational field. Each transition requires bilateral synaptic reconfiguration. Cost accumulates continuously during waking. *Geometric signature:* High behavioural flexibility. Multiple accessible basins with low to moderate barriers. High sleep, typically consolidated bilateral deep sleep. *Examples:* Felidae (domestic cat navigating wild/domestic dual basin), Basin Ω organisms.
2. **Axis 2 — Within-basin metabolic cost.** A single highly expensive basin consumes all available metabolic resources during waking. Sleep is required not for synaptic reconfiguration but for metabolic resource conservation and recovery. *Geometric signature:* Low behavioural flexibility. Single basin. Complete basin commitment. High sleep from resource depletion, not navigational cost. *Example:* Koala.
3. **Axis 3 — Within-basin computational intensity.** Extreme neural computation within a single highly specialised sensorimotor basin during waking. Synaptic reconfiguration cost from processing intensity, not switching. *Geometric signature:* Low behavioural flexibility. High performance within a single specialised basin. High sleep from computational cost, not navigational switching. *Example:* Brown bat (echolocation-guided aerial interception at millisecond resolution for 4 hours requires 20 hours of synaptic reset).

The Lock organism operates on none of these axes. Zero navigational demand. Zero within-basin metabolic surplus cost. Zero within-basin computational intensity. The sleep minimum is the signature of zero cost at every axis simultaneously.

3. The Sleep Topologies

The framework predicts that different landscape positions produce not just different sleep durations but different sleep architectures — different topologies of the discharge process. These topologies are geometric predictions.

TOPOLOGY 1: Metabolic quiescence / biochemical rhythm only

Landscape position: Deepest single basin. No nervous system or minimal neural architecture. Zero navigational demand. Programme executes without reconfiguration.

Discharge process: No sleep in any neurological sense. Circadian biochemical rhythms regulate metabolic processes only. No synaptic cost to discharge because no synapses that reconfigure.

Duration: Undefined or approaching zero.

Examples: Sponge, jellyfish (most species), horseshoe crab.

Geometric prediction: This topology is the signature of the deepest possible basin. Any organism showing this topology is at or near the maximum perturbation-invariance extreme of the Lock strategy.

TOPOLOGY 2: Distributed bilateral rest — the ectotherm Navigator solution

Landscape position: Moderate Navigator. Multiple accessible basins with moderate barriers. Moderate navigational cost. But ectothermic metabolic architecture means the energy budget for full bilateral wakefulness is limited to a short window per day.

Discharge process: True bilateral deep sleep (EEG-confirmed) for 2–6 hours. Extended distributed bilateral rest state (reduced neural activity, bilateral EEG, one eye open behavioural vigilance maintained through brainstem-level alerting, not cortical hemispheric asymmetry) for 12–16 hours. Total Navigator-equivalent discharge delivered across the full inactive period rather than concentrated into a single window.

Why bilateral and not unihemispheric? Because the navigational cost is moderate and does not require hemispheric asymmetry in its discharge. The switching cost is between ecological basins, not between waking hemispheres. Both hemispheres carry the cost equally and both discharge it simultaneously across the extended rest period.

Duration: 2–6h true sleep plus 12–16h Basin C rest. Total equivalent: Navigator range.

Example: Crocodilian. Confirmed bilateral EEG during rest periods (Kelly & Lesku, 2015, 2019). One-eye-open vigilance is behavioural, not unihemispheric in the neurological sense. This is the critical

distinction.

Geometric prediction: This topology will be found specifically in ectothermic Navigators — organisms with ecological generalism whose metabolic architecture precludes concentrated bilateral deep sleep windows. It will not be found in Lock organisms (too little cost to distribute) or in endothermic Navigators (sufficient metabolic budget for concentrated sleep). The crocodilian is the confirmed case. Others predicted: large ectothermic generalist reptiles, some large ectothermic fish.

TOPOLOGY 3: Consolidated bilateral deep sleep

Landscape position: Moderate to high navigational cost. Endothermic metabolic architecture with sufficient budget for full bilateral wakefulness followed by concentrated sleep window. Single or dual basin landscape.

Discharge process: Concentrated bilateral deep sleep (NREM and REM) within a single extended window. Full hemispheric offline state. Full synaptic downscaling.

Duration: 7–22 hours depending on axis and intensity.

Examples: Rodent, human, koala, brown bat, lion. The standard mammalian and avian sleep architecture. Most of the published comparative sleep literature describes organisms in this topology.

Geometric prediction: This topology is the signature of endothermic organisms. Within this topology, duration discriminates between Axis 1 (switching cost), Axis 2 (metabolic cost), Axis 3 (computational cost), and the Lock minimum (zero cost). Duration is a valid direct proxy for total cost within this topology only.

TOPOLOGY 4: Permanent Basin Ω — dissolved barrier

Landscape position: High navigational cost in an environment where full bilateral sleep is lethal. The cost of the sleep basin's exit barrier has been inverted — entering full bilateral sleep has higher mortality cost than the accumulated synaptic cost of remaining partially awake indefinitely.

Discharge process: The barrier between waking and sleep is dissolved. One hemisphere sleeps while the other remains active, in continuous alternation. The organism permanently occupies Basin Ω of the waking/sleep landscape. Discharge is continuous and partial rather than periodic and complete.

Why unihemispheric and not bilateral? Because the discharge must maintain motor and sensory function on the non-sleeping side at all times. This is a constraint imposed by the landscape (open ocean mandatory ventilation plus multi-directional predation threat), not by the navigational cost itself. The navigational cost could in principle be discharged bilaterally if the environment permitted it. It cannot. Unihemispheric is the only geometrically feasible topology for this landscape position.

Duration: Approximately 8 hours total but never bilateral. Continuous rotation.

Examples: Dolphin, porpoise, some seals. Some migratory birds during flight.

Geometric prediction: This topology will be found exclusively in organisms whose landscape position combines high navigational cost with a lethal cost attached to full bilateral sleep. It requires both conditions simultaneously. Neither alone produces the topology.

TOPOLOGY 5: Fragmented shallow sleep — the shallow basin signature

Landscape position: Deep single basin but with high external pressure (predation, ecological constraint) making the sleep basin difficult to occupy for more than short bursts. The sleep basin is not shallow due to low cost — it is shallow because the waking basin is maintained under high external pressure that continually pulls the system back before deep sleep can complete.

Discharge process: Brief fragmented bilateral sleep bouts. Incomplete discharge per bout. Discharge accumulated across many short sessions.

Duration: 30 minutes to 2 hours total, distributed across the day.

Examples: Giraffe. Migratory birds during peak migration (Basin A so deep that sleep is deferred entirely until migration terminates, then discharged as sleep rebound).

Geometric prediction: Sleep rebound is the diagnostic signature of this topology. An organism in Topology 5

that is temporarily relieved of the external pressure will show immediate and disproportionate sleep increase — paying accumulated deferred debt. This distinguishes it from a Lock organism (which will not show sleep rebound under safety conditions because it has no accumulated debt) and from an endothermic Navigator (which discharges continuously and has no large debt to rebound from).

The five topologies are not a taxonomy of sleep types. They are the geometric predictions of the framework applied to the waking/sleep sub-landscape within G . Each topology is the only architecturally feasible solution for its corresponding landscape position. The framework does not say organisms “choose” a sleep topology. It says the landscape position geometrically determines which topology is accessible.

4. The Two Survival Strategies

The deep-time survival data reveals two geometrically opposite strategies that both produce long-term lineage persistence.

4.1. Strategy 1: The Lock

Basin structure	Single basin. Extremely deep. Very high exit barrier.
Morphology	Extreme conservatism. Body plan unchanged across hundreds of millions of years.
Metabolism	Slow. Efficient. Minimal resource requirement.
Behaviour	Low flexibility. Narrow repertoire. Highly programmatic.
Sleep quantity	Minimal. Approaching zero in simplest forms. 2–5 hours in complex Lock organisms.
Sleep topology	Topology 1 (quiescence/biochemical rhythm) in simplest forms. Topology 5 (fragmented, predation pressure) in complex Lock organisms with environmental threat. Never Topology 4 (no open-ocean lethal constraint). Never Topology 2 (insufficient navigational cost to distribute).
Survival horizon	Maximum. 300–600 Myr+.
Survival mechanism	Basin so deep it is perturbation-invariant. Extinction events cannot displace the organism.
Convergence	Multiple independent phyla.

4.2. Strategy 2: The Navigator

Basin structure	Multiple basins. Shallow barriers. High switching capacity.
Morphology	Ecological generalism. Body plan adapted for flexibility.
Metabolism	Flexible. Moderate. Endothermic or ectothermic — but the metabolic architecture determines which sleep topology is available to discharge the Navigator cost.
Behaviour	High flexibility. Broad repertoire. Contextual switching.
Sleep quantity	Moderate equivalent. The total Navigator-range discharge cost is constant across Navigators but is delivered through different topologies depending on metabolic architecture.
Sleep topology	Endothermic Navigator: Topology 3 (consolidated bilateral deep sleep, 8–14h). Ectothermic Navigator: Topology 2 (distributed bilateral rest, 2–6h true sleep + 12–16h Basin C rest, total equivalent Navigator range). Open-ocean Navigator with lethal bilateral constraint: Topology 4 (permanent Basin Ω , unihemispheric alternation).
Survival horizon	Long. 100–300 Myr.
Survival mechanism	Navigates to alternative basin in real time when current basin disappears. Generalism is itself Basin Ω of ecological flexibility.
Convergence	Multiple independent lineages.

4.3. The Unstable Zone

Basin structure	Shallow single basin. High barriers to alternative basins.
Morphology	Narrow specialist. Not deep enough to be perturbation-invariant. Not flexible enough to reach alternative basins when niche disappears.
Sleep quantity	Variable. Often high (Axis 2 or 3) but with no survival advantage.
Sleep topology	Typically Topology 3 (consolidated bilateral) but the duration reflects niche-maintenance cost, not navigational capacity. The topology does not discriminate Unstable Zone organisms from Navigator organisms in duration alone — the discriminator is behavioural flexibility and dietary breadth, not sleep architecture.
Survival horizon	Short. Millions of years.
Examples	<i>T. rex</i> , mammoth, sabre-tooth cat, passenger pigeon, dodo.

4.4. Why The Lock Outlasts The Navigator

The Lock survives longer than the Navigator for a precise geometric reason. The Navigator requires accessible alternative basins at the moment of disruption. Catastrophic perturbations that eliminate all accessible basins simultaneously defeat the Navigator. The Lock cannot be displaced because its basin depth exceeds the perturbation energy of any event the biosphere has yet produced.

The Permian–Triassic event killed 96% of all marine species. The horseshoe crab was still there on the other side.

Basin depth is the ultimate survival parameter. Navigation is a survival strategy for moderate perturbations. For catastrophic perturbations, only the deepest basins survive. The minimum of the sleep spectrum is the maximum of the survival spectrum. The topology of that minimum is the geometry of a basin so deep it needs no maintenance.

5. The Empirical Record

5.1. Lock Strategy: Confirmed Examples

Lineage	Age	Sleep quantity	Sleep topology	Basin geometry
Sponge (Porifera)	~600 Myr	Undefined. No nervous system.	Topology 1. Circadian biochemical only.	Deepest possible. Sessile. Single programme.
Jellyfish (Cnidaria)	500–700 Myr	Undefined for most. <i>Cassiopeia</i> rest ~8h.	Topology 1 for most. Topology 5 fragment in <i>Cassiopeia</i> .	No brain. Radial symmetry. Passive drift and sting.
Nautilus (Mollusca)	~500 Myr	No REM. Quiescent periods distributed.	Topology 1 to Topology 2 border. Bilateral rest. No hemispheric architecture.	Neutral buoyancy. Passive ambush. Shell unchanged.
Horseshoe crab (Chelicerata)	~450 Myr	Tidal/circadian rest only.	Topology 1. No bilateral sleep.	Deepest terrestrial Lock. 450 Myr unchanged.
Shark (Chondrichthyes)	~400 Myr	No true bilateral sleep. Rest periods ~2–4h distributed.	Topology 2 border. Distributed rest. No REM confirmed.	Cartilage. Generalist enough to border Lock/Navigator by species.
Coelacanth (Sarcopterygii)	~400 Myr	Unknown. Presumed minimal.	Topology 1 to 2. Deep cold water suppresses all activity.	Minimal metabolism. Unchanged 400 Myr.
Tuatara (Lepidodosauria)	~200 Myr	12h+ rest. Most is torpor-like low-temp dormancy.	Topology 2. Distributed ectotherm rest. Not true bilateral deep sleep.	Slowest reptile metabolism. Functions at 5°C. Lives 100+ years.

5.2. Navigator Strategy: Confirmed Examples

Lineage	Age	Sleep quantity	Sleep topology	Basin geometry
Cockroach (Blattodea)	~300 Myr	~12h rest cycle.	Topology 2. Distributed bilateral rest. Ectotherm.	Eats anything. Lives anywhere. 300 Myr survival.
Crocodylian (Archosauria)	~200 Myr	2–6h true bilateral EEG sleep. 12–16h Basin C distributed bilateral rest. Total equivalent: Navigator range.	Topology 2. Bilateral EEG throughout (Kelly & Lesku, 2015, 2019). One-eye-open vigilance is <i>behavioural</i> , not unihemispheric in the neurological sense. The predicted topology for an ectothermic Navigator.	Generalist predator. Survived K–Pg. Fresh/salt water. Any prey. 200 Myr.
Dolphin (Cetacea)	~50 Myr	~8h total. Never bilateral.	Topology 4. Permanent Basin Ω . Unihemispheric alternation. Waking/sleep barrier dissolved.	Open ocean. High cognitive complexity. Must surface to breathe. Bilateral sleep lethal.
Rodent (Muridae)	~65 Myr	~13h.	Topology 3. Consolidated bilateral deep sleep. Standard endothermic Navigator.	Extreme generalist. All biomes and diets.
Corvidae	~30 Myr	~8–10h.	Topology 3. Standard avian bilateral. Some Topology 4 during roosting.	Problem solving. Tool use. Social complexity.

5.3. Unstable Zone: Confirmed Examples

Lineage	Age	Sleep quantity	Sleep topology	Why extinct
<i>T. rex</i> (Tyrannosauridae)	2–3 Myr	Unknown. Moderate large predator presumed.	Topology 3 presumed.	Narrow apex predator. No alternative basin at K–Pg.
Mammoth (Proboscidea)	0.5–5 Myr	Likely ~2h (cf. elephant).	Topology 5 (size + predation pressure).	Grassland specialist. Habitat contracted. Could not adapt.
Sabre-tooth cat (Machairodontinae)	0.5–11 Myr	High felid-range likely.	Topology 3. Basin Ω felid ancestry.	Specialist predator. Megafauna gone. Basin collapsed.
Dodo (Raphidae)	0.003 Myr	Standard avian ~8–10h.	Topology 3.	Island specialist. Single shallow basin. First perturbation = extinction.

6. The Sleep Spectrum as Temporal Sonar

6.1. The Full Sonar Instrument

The sleep sonar has two channels, not one. Duration alone is insufficient. Duration and topology together triangulate the organism’s position in the landscape with full geometric precision.

Duration	Topology	Survival horizon	Landscape position	Examples
~0	T1	600 Myr+	Deepest Lock. No nervous system. Zero cost, zero discharge.	Sponge, jellyfish
0–4h	T1–T2	300–500 Myr	Deep Lock. Minimal neural architecture. Programme runs without maintenance.	Horseshoe crab, shark, coelacanth
2–6h true + T2 rest	T2	100–300 Myr	Ectothermic Navigator. Navigator cost distributed across extended bilateral rest. Total equivalent: Navigator range.	Crocodylian, cockroach, tuatara
8–14h concentrated	T3	10–100 Myr	Endothermic Navigator or coupled social field. Standard bilateral discharge.	Rodent, corvid, dog, human
8h, never bilateral	T4	50–200 Myr	Open-ocean Navigator. Lethal bilateral constraint. Basin Ω of waking/sleep landscape.	Dolphin
0.5–2h fragmented	T5	Variable — external pressure dependent	Single deep basin under strong external pressure. Sleep basin shallow by external force, not low cost. Sleep rebound diagnostic.	Giraffe
14–22h	T3	Niche-dependent	Axis 2 (metabolic), Axis 3 (computational), or deep Basin Ω (Felidae).	Koala, bat, lion, cat

6.2. The U-Shaped Survival–Duration Relationship

The survival–sleep duration relationship is U-shaped with a flat minimum. The minimum of the sleep duration spectrum corresponds to the maximum of the survival spectrum (Lock strategy). The moderate range is the Navigator survival zone. The high-duration range is the uncertain zone — survival depends on niche stability or Basin Ω depth.

Critical note on duration alone: Duration without topology is ambiguous. A crocodylian sleeping 2–6 hours of true bilateral sleep appears to be in the Lock range. It is not. It is an ectothermic Navigator using Topology 2 to distribute Navigator-range cost. The topology disambiguates.

Similarly, a giraffe sleeping 30 minutes appears to be a deep Lock. It may be. But the diagnostic is sleep rebound under safe conditions. If the giraffe increases sleep substantially when predation threat is removed, it is a Topology 5 organism with accumulated debt, not a Lock. If it does not increase sleep under safety, the basin depth is geometric and the Lock classification holds.

Duration + Topology + Rebound response: these three together uniquely specify the organism’s landscape position. None of the three alone is sufficient.

7. Convergence Across Independent Phyla

7.1. The Lock Converges Independently

Phylum / Class	Example	Age	Sleep topology
Porifera	Sponge	~600 Myr	T1
Cnidaria	Jellyfish	500–700 Myr	T1
Mollusca	Nautilus	~500 Myr	T1–T2 border
Chelicerata	Horseshoe crab	~450 Myr	T1
Chondrichthyes	Shark	~400 Myr	T2 border
Sarcopterygii	Coelacanth	~400 Myr	T1–T2
Lepidosauria	Tuatara	~200 Myr	T2

Seven independent phyla. Seven independent arrivals at the same geometric strategy. The Lock topology is Topology 1 at the extreme and transitions toward Topology 2 as neural complexity increases while remaining in a single deep basin. This transition within the Lock is itself a geometric prediction: as Lock organisms develop more complex nervous systems, they acquire more synaptic cost to discharge, but the basin remains deep enough to keep total cost minimal. Topology shifts from T1 to T2 but duration remains in the Lock range.

7.2. The Navigator Converges Independently, Across Multiple Topologies

Lineage	Example	Age	Sleep topology
Blattodea	Cockroach	~300 Myr	T2 (ectotherm)
Archosauria	Crocodylian	~200 Myr	T2 (ectotherm, confirmed bilateral EEG)
Cephalopoda	Octopus	~300 Myr	T2–T3 border (distinct active/quiet sleep phases confirmed)
Cetacea	Dolphin	~50 Myr	T4 (open ocean lethal bilateral constraint)
Rodentia	Rat	~65 Myr	T3 (endotherm)
Aves	Corvid	~30 Myr	T3, T4 during roosting
Primates	Human	~6 Myr	T3 (endotherm, inter-generational Basin B* coupling)

The critical geometric observation: the Navigator strategy has converged independently across at least three distinct sleep topologies (T2, T3, T4). The strategy is the same — ecological generalism as Basin Ω of ecological flexibility. The topology varies with the metabolic and environmental constraints of the specific lineage. This is exactly what the framework predicts: the topology is determined by the sub-constraints of the landscape position, not by the strategy classification.

The Navigator strategy is the deep basin. The topology is the solution the landscape provides for paying the cost of occupying that basin given the full constraint set.

7.3. The Unstable Zone Does Not Converge

Mass extinction casualties do not converge. Each extinction event eliminates different taxa via different mechanisms. The unstable zone has no deep basin by definition. The only convergent feature of the unstable zone is its instability. Topology does not discriminate here: Unstable Zone organisms occupy Topology 3 (same as endothermic Navigators) but lack the dietary breadth and behavioural flexibility that makes the Navigator strategy viable. Duration and topology together still fail to distinguish Unstable Zone from Navigator without the behavioural flexibility measurement — which is why the full three-channel sonar (duration + topology + flexibility) is the complete instrument.

8. The Human Position

The human is an endothermic Navigator with Topology 3 sleep (consolidated bilateral deep sleep, 7–9 hours) and an unprecedented inter-generational Basin B* — a co-constructed social attractor that persists beyond the individual lifetime via cultural transmission.

The human sleep topology (T3) is standard endothermic Navigator. The sleep duration (7–9 hours) is standard Navigator range. Nothing in the human’s sleep signature is anomalous with respect to the framework.

What is anomalous is the Basin B* depth increasing across generations through cultural accumulation. This creates a new geometric feature not present in any other known organism: a basin whose depth is a function of time and investment rather than purely of evolutionary selection. The framework does not yet predict the topological consequences of a basin that deepens in real time at inter-generational scale. This is the honest limit of the current derivation.

The risk: the same cultural Basin B* is remodelling the ecological landscape faster than any prior organism. A Navigator survives by finding alternative basins when the current one disappears. If the Navigator is simultaneously the agent collapsing the alternative basins for all other species, the Navigator strategy fails even for the Navigator. This is a geometric statement, not a moral one.

9. Literature Confirmations

9.1. Oldest Surviving Lineages Sleep the Least

Derived before literature: Minimum cost = minimum discharge = maximum basin depth = maximum survival.

Confirmed: Horseshoe crab, nautilus, jellyfish, shark. All Lock organisms confirmed as minimal discharge. All show Topology 1 or Topology 2. Source: Comparative sleep literature; living fossil biology.

9.2. Crocodilian Sleep is Bilateral EEG, Not Unihemispheric

Derived before literature: Ectothermic Navigator topology prediction — distributed bilateral rest, not unihemispheric.

Confirmed: Kelly & Lesku (2015) EEG study of juvenile *Crocodylus niloticus*: bilateral EEG throughout rest periods. One-eye-open vigilance is behavioural, maintained via brainstem alerting, not cortical hemispheric asymmetry. This is the predicted T2 topology for an ectothermic Navigator. Source: Kelly, M.L. & Lesku, J.A. (2015), *J. Exp. Biol.*, 218, 3175–3180; Kelly & Lesku (2019), *Biol. Rev.*

9.3. Mass Extinction Survivors Are Disproportionately Lock or Navigator

Derived: Unstable zone is where extinction concentrates casualties.

Confirmed: K–Pg survivors: crocodilians (Navigator, T2), turtles (partial Lock, T2), birds (Navigator, T3 and T4), small mammals (Navigator, T3). Permian–Triassic survivors: horseshoe crab (Lock, T1), nautilus (Lock, T1). Source: Jablonski (1986); Barnosky et al. (2011).

9.4. Both Strategies Convergent Across Independent Phyla

Derived: Lock and Navigator are deep basins in evolutionary strategy space, therefore convergent.

Confirmed: Convergence confirmed in phyla listed in Section 7. Source: Conway Morris (2003, *Life's Solution*).

9.5. Slow Metabolism Correlates With Deep-Time Survival

Derived: Lock strategy requires minimal metabolic expenditure.

Confirmed: Tuatara, coelacanth, horseshoe crab. All show Lock topology (T1–T2) and slow metabolism as convergent traits. Source: Comparative metabolism literature; Cree (2014) tuatara biology.

9.6. Generalist Diet and Ecological Flexibility Predict Extinction Survival

Derived: Navigator mechanism is generalism as Basin Ω .

Confirmed: Capellini et al. (2008, *Proc Royal Soc B*); Barnosky et al. (2011).

10. The Novel Predictions

All predictions stated before data gathered to test them. Locked here. 2026-03-18.

10.1. P-INVARIANT-1: Sleep Duration Negatively Correlates With Lineage Age Across All Phyla, Controlling For Topology

A cross-phyla analysis of sleep duration (or rest-state equivalent, topology- calibrated) against lineage age will show a negative correlation surviving correction for body mass, metabolic rate, and topology class. The oldest lineages will have the lowest topology-calibrated discharge requirement. Detectable across Porifera, Cnidaria, Mollusca, Arthropoda, and Vertebrata simultaneously.

Discriminating value: This prediction is not generated by any existing sleep function theory. It requires both the dimensional invariant and the topology-calibration framework.

10.2. P-INVARIANT-2: Mass Extinction Survivor Lineages Sleep Less (Lock) or Show Higher Dietary Breadth (Navigator) Than Casualty Sister Taxa

For each major mass extinction event, Lock survivors sleep less than casualties. Navigator survivors show higher dietary breadth than casualties and occupy T2, T3, or T4 (not T5 under chronic ecological pressure). Testable via fossil record body plan conservatism proxy, dietary isotope and dental analysis, and phylogenetic comparison.

10.3. P-INVARIANT-3: The Unstable Zone Boundary Is Predictable From Duration, Topology, and Dietary Breadth

A three-dimensional boundary in the space of (sleep duration, sleep topology, dietary breadth) separates Lock, Navigator, and Unstable Zone organisms. The two-dimensional boundary (duration $\times$ dietary breadth) of v1.0 is confirmed but the addition of topology as the third axis removes the ambiguity between ectothermic Navigators (T2, appearing Lock-range in duration alone) and true Lock organisms.

10.4. P-INVARIANT-4: Topology 2 Will Be Found Specifically and Exclusively in Ectothermic Navigators

Topology 2 (distributed bilateral rest with no EEG hemispheric asymmetry) will be found in organisms satisfying both: (a) ecological generalism (Navigator strategy), and (b) ectothermic metabolic architecture. It will not be found in Lock organisms (insufficient navigational cost to distribute) or in endothermic Navigators (sufficient metabolic budget for concentrated sleep). The crocodilian is the confirmed case. Predicted additionally: large ectothermic generalist reptiles (e.g., monitor lizards, large iguana species), and large ectothermic generalist fish.

10.5. P-INVARIANT-5: Sleep Rebound Uniquely Identifies Topology 5 And Distinguishes It From Lock

Organisms showing minimal sleep (30 minutes to 2 hours) will divide into two classes under safe-condition testing: *Class A (Lock)*: No significant sleep increase when predation pressure is removed. No rebound. Basin is geometrically deep. Minimal sleep is the geometric minimum, not a deferred debt. *Class B (Topology 5)*: Significant sleep increase when pressure is removed. Rebound present. Sleep was suppressed by external pressure, not absent by geometric necessity. The giraffe is the predicted Class B case. Migratory birds post-migration are the confirmed Class B case.

10.6. P-INVARIANT-6: Lock Organisms Are Undomesticable

The same basin depth that produces perturbation-invariance in the face of mass extinctions renders the basin immovable by any domestication field that does not also destroy the organism. No Lock organism (T1 or T2 topology, minimal duration, Lock-range dietary breadth) has been domesticated. This will remain true.

10.7. P-INVARIANT-7: The Dimensional Invariant Extends Below Nervous System to Metabolic Rest State Duration in Single-Celled Organisms

Single-celled organisms show metabolic rest states whose duration tracks niche breadth in the same direction the invariant predicts. Specialist bacteria (narrow niche, deep metabolic basin) show longer continuous activity with minimal rest. Generalist bacteria show moderate distributed rest. This extends the invariant below the level of neural architecture entirely — confirming it as a property of basin topology, not of sleep as a neurological phenomenon.

11. The Dimensional Invariant: Complete Formal Statement

THE DIMENSIONAL INVARIANT: COMPLETE STATEMENT

Across all life, at all scales, across all of deep time, the framework $S + N + G \rightarrow R$ makes two inseparable predictions about rest:

1. Quantity: The total cost of basin occupation (decomposed into Axis 1 between-basin switching cost, Axis 2 within-basin metabolic cost, and Axis 3 within-basin computational intensity) must be discharged per cycle. Duration of discharge tracks total cost. Basin depth — resistance to perturbation — is the primary predictor of survival. The oldest surviving lineages have the lowest basin-occupation cost. They discharge the least.

2. Topology: The architecture of the discharge process — the sleep topology — is a further geometric prediction of the same framework. The landscape position determines not only how much must be discharged but what architecture of discharge is geometrically feasible given the full constraint set (metabolic architecture, environmental lethality of full sleep, navigational demand structure).

Five topologies are predicted: T1 (quiescence/biochemical rhythm) for the deepest Lock organisms; T2 (distributed bilateral rest) for ectothermic Navigators; T3 (consolidated bilateral deep sleep) for endothermic organisms at all cost levels; T4 (permanent Basin Ω , unihemispheric) for organisms where full bilateral sleep is lethal; T5 (fragmented shallow sleep with rebound) for organisms under chronic external suppression of sleep.

The invariant holds because basin depth is the same geometric property at every scale. Sleep — in whatever topology the landscape geometry permits — is the payment. Survival is the depth.

The oldest surviving lineages sleep the least.
How they sleep is the geometry of where they are.
This is not a coincidence.
This is the geometry.

12. The Unified Derivation Chain

Step	Claim	Record
1. Cancer	FOXA1/EZH2 ratio identifies whether a cancer cell is epigenetically arrested in the cancer attractor or has lost access to the identity basin.	10.5281/zenodo.18883922
2. Domest- tication	The domestication syndrome is field-specified developmental arrest of NCC migration. The pig is the clean case.	10.5281/zenodo.19094935
3. Canine	Above a genomic fixation threshold, the landscape is remodelled. Basin A becomes inaccessible.	Repository 2026-03-18
4. Feline	The symmetric dual-basin landscape. The three-class domestication taxonomy. Basin Ω as meta-attractor.	Repository 2026-03-18
5. Sleep cost	Sleep duration tracks the total cost of basin occupation. Three axes. Universal principle.	Repository 2026-03-18
6. Sleep sonar	The sleep spectrum maps onto the three-axis framework across all animals.	Repository 2026-03-18
7. This document	The dimensional invariant. Sleep quantity and sleep topology are both geometric predictions of the landscape. The oldest surviving lineages sleep the least. The topology of how they sleep is the geometry of where they are.	This document

THE UNIFIED STATEMENT

The attractor geometry framework is a single geometric principle operating identically at every scale at which life organises itself.

From a cancer cell deciding whether to differentiate or proliferate, to a domestic pig deciding whether to feral, to a cat choosing which basin to occupy this morning, to a crocodile distributed across 16 hours of bilateral rest because its metabolic budget cannot afford the concentrated sleep that its navigational cost demands, to a dolphin that has dissolved the barrier between waking and sleep and now permanently inhabits the intermediate, to a horseshoe crab that has been in the same basin for 450 million years and does not sleep at all because there is nothing to reset:

the geometry is the same.
The basin is the same concept.
The cost of navigation is the same cost.
The sleep is the same payment.
The topology of sleep is the shape of the landscape.

The survival is the depth.
Scale is the only variable.
The geometry is invariant.

Linked Records

Record	DOI / Location
Theory paper	doi:10.5281/zenodo.19094935
P7 pre-registration	doi:10.5281/zenodo.19096128
P9 pre-registration	[DOI pending — insert after confirmation]
FOXA1/EZH2 cancer	doi:10.5281/zenodo.18883922
Plant inversion	doi:10.5281/zenodo.19040399
LECA resurrection	doi:10.5281/zenodo.18986790
Canine / feline / sleep artifacts	github.com/Eric-Robert-Lawson/ attractor-oncology (2026-03-18)

Document ID: DIMENSIONAL_INVARIANT_SLEEP_SURVIVAL_REASONING_ARTIFACT_v2.0
Date: 2026-03-18
Author: Eric Robert Lawson / OrganismCore
Status: REASONING ARTIFACT — DERIVATION RECORD. Version 2.0 extends v1.0 by making explicit the topology dimension as a geometric prediction of the framework, not a measurement correction. Primary theoretical contributions: (1) The dimensional invariant (oldest surviving lineages sleep the least; minimum sleep = maximum survival horizon; U-shaped survival-sleep relationship with flat minimum). (2) The two-strategy taxonomy (Lock and Navigator). (3) The unstable zone identification. (4) The five sleep topologies as geometric predictions of the landscape, not empirical sleep types. (5) The full three-channel sonar (duration + topology + flexibility). (6) Predictions P-INVARIANT-1 through P-INVARIANT-7.
Theory paper: [doi:10.5281/zenodo.19094935](https://doi.org/10.5281/zenodo.19094935)
Repository: [github.com/Eric-Robert-Lawson/ attractor-oncology](https://github.com/Eric-Robert-Lawson/attractor-oncology)
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